LAB SHEET 08

TEMPLATE INHERITANCE, BLUEPRINTS & MODULAR UI ENGINEERING

Primary Syllabus Exercise Reference:

Develop a core master template layout blueprint file containing shared menus and block overrides; inherit properties down across specialized views (Home, About Us, Contact Us).

Task 8.1 – Dynamic Active Navigation Highlight

```
from flask import Flask
from main.routes import main_bp

app = Flask(__name__)
app.secret_key = "lab-sheet-08-demo-key"

app.register_blueprint(main_bp)

if __name__ == "__main__":
    app.run(debug=True)
from flask import Blueprint

main_bp = Blueprint("main", __name__)

from . import routes
from flask import render_template, request, redirect, url_for, flash, current_app

from . import main_bp

@main_bp.get("/")
def home():
    return render_template("home.html", active_page="home")

@main_bp.get("/about")
def about():
    return render_template("about.html", active_page="about")

@main_bp.route("/contact", methods=["GET", "POST"])
def contact():
    if request.method == "POST":
        feedback = request.form.get("feedback", "").strip()

        if feedback:
            current_app.logger.info(
                "Verified Contact Feedback: %s", feedback
            )
            flash("Thank you! Your feedback was received.", "success")
        else:
            flash("Please enter your feedback before submitting.", "error")

        return redirect(url_for("main.contact"))

    return render_template("contact.html", active_page="contact")
<!doctype html>
```

```
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>{% block title %}Modular Web App{% endblock %}</title>
   <link rel="stylesheet"
        href="{{ url_for('static', filename='style.css') }}">
</head>
<body>

<nav class="navbar">
   <a class="{% if active_page == 'home' %}active{% endif %}"
     href="{{ url_for('main.home') }}">Home</a>

   <a class="{% if active_page == 'about' %}active{% endif %}"
     href="{{ url_for('main.about') }}">About Us</a>

   <a class="{% if active_page == 'contact' %}active{% endif %}"
     href="{{ url_for('main.contact') }}">Contact Us</a>
</nav>

{% block messages %}{% endblock %}

<main class="container">
   {% block content %}{% endblock %}
</main>

</body>
</html>
.navbar {
   display: flex;
   gap: 24px;
   padding: 16px 30px;
   background: #1f2937;
}

.navbar a {
   color: white;
   text-decoration: none;
   padding: 8px 14px;
   border-radius: 6px;
}

.navbar a.active {
   background: #2563eb;
   font-weight: 700;
}
{% extends "base.html" %}

{% block title %}Home{% endblock %}

{% block content %}
<h1>Welcome to the Home Page</h1>
<p>This page inherits the master layout.</p>
{% endblock %}
```

Expected Output

Browser Output – Home

[Home] About Us Contact Us

Welcome to the Home Page
This page inherits the master layout.

The Home navigation item is highlighted.

```
{% extends "base.html" %}

{% block title %}About Us{% endblock %}

{% block content %}
<h1>About Us</h1>
<p>We build modular and reusable web interfaces.</p>
{% endblock %}
```

Browser Output – About Us

Home [About Us] Contact Us

About Us
We build modular and reusable web interfaces.

The About Us navigation item is highlighted.

Task 8.2 – Persistent User Feedback Notification

```
{% with messages = get_flashed_messages(with_categories=true) %}
  {% if messages %}
    <section class="flash-area">
      {% for category, message in messages %}
        <div class="alert {{ category }}">
          {{ message }}
        </div>
      {% endfor %}
    </section>
  {% endif %}
{% endwith %}

{% block content %}{% endblock %}
.flash-area {
  max-width: 900px;
  margin: 18px auto;
}

.alert {
  padding: 12px 16px;
  border-radius: 7px;
  margin-bottom: 10px;
  font-weight: 600;
}

.alert.success {
  background: #dcfce7;
  border: 1px solid #86efac;
}
```

```
.alert.error {
    background: #fee2e2;
    border: 1px solid #fca5a5;
}
```

Expected Output

After successful Contact Us submission:

✓ Thank you! Your feedback was received.

The notification appears through the inherited base.html
message hook and is available consistently across child pages.

Task 8.3 – Professional Contact Us Form & Server Logging

```
{% extends "base.html" %}

{% block title %}Contact Us{% endblock %}

{% block content %}
<div class="contact-card">
    <h1>Contact Us</h1>
    <p>We would love to hear your feedback.</p>

    <form method="POST" action="{{ url_for('main.contact') }}">
        <label for="feedback">Your Feedback</label>

        <textarea id="feedback"
                  name="feedback"
                  rows="7"
                  placeholder="Enter your feedback here..."
                  required></textarea>

        <button type="submit">Send Feedback</button>
    </form>
</div>
{% endblock %}
```

```
.contact-card {
    max-width: 720px;
    margin: 40px auto;
    padding: 30px;
    background: white;
    border-radius: 12px;
    box-shadow: 0 8px 24px rgba(0,0,0,.08);
}

.contact-card h1 {
    margin-top: 0;
}

.contact-card label {
    display: block;
    margin: 20px 0 8px;
    font-weight: 700;
}
```

```
.contact-card textarea {
    width: 100%;
    box-sizing: border-box;
    padding: 12px;
    border: 1px solid #cbd5e1;
    border-radius: 7px;
    resize: vertical;
}

.contact-card button {
    margin-top: 16px;
    padding: 11px 20px;
    border: 0;
    border-radius: 7px;
    background: #2563eb;
    color: white;
    font-weight: 700;
    cursor: pointer;
}
```

Expected Output

```
Browser Output – Contact Us


-----------------------------------------------------
Contact Us

We would love to hear your feedback.

Your Feedback
[ Enter your feedback here...                ]
[                                            ]
[                                            ]

                      [ Send Feedback ]
-----------------------------------------------------
```

```
<!doctype html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>{% block title %}Modular Web App{% endblock %}</title>
    <link rel="stylesheet"
          href="{{ url_for('static', filename='style.css') }}">
</head>
<body>

<nav class="navbar">
    <a class="{% if active_page == 'home' %}active{% endif %}"
       href="{{ url_for('main.home') }}">Home</a>
    <a class="{% if active_page == 'about' %}active{% endif %}"
       href="{{ url_for('main.about') }}">About Us</a>
    <a class="{% if active_page == 'contact' %}active{% endif %}"
       href="{{ url_for('main.contact') }}">Contact Us</a>
</nav>

{% with messages = get_flashed_messages(with_categories=true) %}
    {% if messages %}
```

```
        <section class="flash-area">
            {% for category, message in messages %}
                <div class="alert {{ category }}">
                    {{ message }}
                </div>
            {% endfor %}
        </section>
    {% endif %}
{% endwith %}

<main class="container">
    {% block content %}{% endblock %}
</main>

</body>
</html>
```